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Roll No

EX/EE-7003-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Computer Application to Power System

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Write a brief note on line loadability and what do you understand by capability curves of alternator.
b) Define Tree, Co-tree and Branch. Also explain with a simple example why Nodal analysis is preferred over Mesh analysis.
2. a) What components of power system absorber generate reactive power? How they can be quantized?
b) Discuss the method of voltage control in transmission system utilizing OLTC transformer.
3. a) What do you understand by regulated shunt compensation?
b) Distinguish between Series and Shunt compensation.
4. a) Explain "Power system security" concept. Enlist various security function and security level.
b) Discuss the method of economic dispatch using LP formulation.

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5.
 - a) Describe voltage stability of power system and discusses their control variable with example.
 - b) Explain the application of sensitivity analysis in power system.
6.
 - a) Discuss the approach for Voltage stability analysis of a given system.
 - b) What are P-V and V-Q curves? Discuss how these help in studying voltage stability of the system.
7.
 - a) Discuss the factor affecting power system security and define Generation Shift Distribution factor.
 - b) Discuss the contingency analysis using network sensitivity method.
8. Write short notes on
 - a) Regulating transformer
 - b) Effect of load models
